# Supplementary Information

OmniGene-4: A Unified Bio-Language MoE Model with Router-Level Interpretability

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## Contents

# 1 Supplementary Note 1: Extended Limitations and Future Work

**Top-1 expert purity is modest.** The highest-purity atomic expert we identified (E54, English function-words at layer 12) reaches 80%, but most "specialty" experts land at 15–35% purity. Our claims about task-specific experts should be understood distributionally rather than as hard assignments.

**Single model family.** All results are from Gemma-4-26B-A4B-Instruct. Whether the 96%/4% CPT/SFT decomposition generalizes to Mixtral, DeepSeek-MoE, or GPT-class bio models is an open question. The architectures differ in expert count, top-$k$ selection, and router objective.

**Remote-homology ceiling.** Our 82.60% remote-homology accuracy leaves an 18% error rate. Specialist methods with structural-alignment heads (ESM-2 + TM-Vec, PLMSearch, CATHe) report 65–75% on differently constructed splits. We hypothesize the remaining gap is a representation-level constraint rather than an SFT-scale constraint; testing requires controlled experiments with more CPT on distant-protein pairs (≈150 GPU-hours, deferred).

**No chain-of-thought reasoning.** We briefly attempted to train with `<THOUGHT>` reasoning blocks for remote homology but abandoned it because synthetic-CoT templates from sequence features were of insufficient quality. Human-expert or strong-teacher-generated CoT data would likely improve remote-homology performance substantially.

**Evaluation scale.** BixBench's True/False subset contains only ≈200 questions; small absolute differences in the 90–95% range are within the dataset's noise floor. Richer evaluation pools (HELM-Bio, LM-BIO) would strengthen the claims.

**No ablation on `router.proj` LoRA.** We state that including `router.proj` in the LoRA target modules is critical to enabling expert re-specialization, based on an informal pilot observation. A controlled ablation — CPT with and without `router.proj` LoRA, holding all other hyperparameters fixed — would quantify this claim.

**Future experiments.** (1) Controlled CPT ablation with richer distant-protein corpora. (2) Router.proj ablation. (3) Human-expert CoT for remote homology. (4) Replication on Mixtral-8x7B or DeepSeek-MoE-16B. (5) Downstream wet-lab validation (variant effect prediction on ClinVar). (6) GGUF quantization for single-GPU deployment on RTX 4090.

# 2 Supplementary Tables

## 2.1 Supplementary Table S1: Full Per-Layer JS Divergence

**Table 1:** Layer-by-layer Jensen–Shannon divergence across training stages. Values are mean pairwise JS across all 28 task pairs at each layer.

| Layer | Baseline | CPT | v3 (CPT+SFT) | Δ CPT |
|---|---|---|---|---|
| 0 | 0.160 | 0.164 | 0.164 | +0.004 |
| 1 | 0.136 | 0.113 | 0.113 | −0.023 |

Table 1 – *Continued*

| Layer | Baseline | CPT | v3 (CPT+SFT) | Δ CPT |
|---|---|---|---|---|
| 2 | 0.172 | 0.164 | 0.164 | −0.008 |
| 3 | 0.151 | 0.134 | 0.134 | −0.017 |
| 4 | 0.119 | 0.113 | 0.113 | −0.006 |
| 5 | 0.159 | 0.160 | 0.160 | +0.001 |
| 6 | 0.131 | 0.111 | 0.111 | −0.020 |
| 7 | 0.129 | 0.123 | 0.123 | −0.006 |
| 8 | 0.134 | 0.126 | 0.126 | −0.008 |
| 9 | 0.174 | 0.153 | 0.153 | −0.021 |
| 10 | 0.197 | 0.159 | 0.159 | −0.038 |
| 11 | 0.160 | 0.143 | 0.143 | −0.017 |
| 12 | 0.182 | 0.149 | 0.149 | −0.033 |
| 13 | 0.143 | 0.127 | 0.127 | −0.016 |
| 14 | 0.157 | 0.137 | 0.137 | −0.020 |
| 15 | 0.198 | 0.170 | 0.170 | −0.028 |
| 16 | 0.139 | 0.121 | 0.121 | −0.018 |
| 17 | 0.184 | 0.166 | 0.166 | −0.018 |
| 18 | 0.148 | 0.134 | 0.134 | −0.014 |
| 19 | 0.124 | 0.123 | 0.123 | −0.001 |
| 20 | 0.140 | 0.130 | 0.130 | −0.010 |
| 21 | 0.136 | 0.119 | 0.119 | −0.017 |
| 22 | 0.162 | 0.149 | 0.149 | −0.013 |
| 23 | 0.157 | 0.138 | 0.138 | −0.019 |
| 24 | 0.141 | 0.122 | 0.122 | −0.019 |
| 25 | 0.180 | 0.149 | 0.149 | −0.031 |
| 26 | 0.161 | 0.152 | 0.152 | −0.009 |
| 27 | 0.178 | 0.172 | 0.172 | −0.006 |
| 28 | 0.217 | 0.197 | 0.197 | −0.020 |
| 29 | 0.240 | 0.222 | 0.222 | −0.018 |

## 2.2 Supplementary Table S2: Bootstrap Confidence Intervals

**Table 2:** Bootstrap 95% confidence intervals for routing metrics (1,000 prompt-level iterations).

| Metric | Mean | 95% CI Lower | 95% CI Upper |
|---|---|---|---|
| Baseline JS | 0.344 | 0.344 | 0.344 |
| CPT JS | 0.440 | 0.440 | 0.441 |
| v3 JS | 0.444 | 0.444 | 0.444 |
| Δ CPT | 0.096 | 0.096 | 0.097 |
| Δ SFT | 0.004 | 0.003 | 0.004 |

## 2.3 Supplementary Table S3: Format-Matched Routing Control

**Table 3:** Layer-averaged JS divergence before and after format matching. Six tasks wrapped in a neutral template (`### Task: {name} / ### Content: {text} / ### End`).

| Condition | JS Divergence | Retention Ratio |
|---|---|---|
| Original (heterogeneous formats) | 0.160 | 100% |
| Format-matched (neutral template) | 0.145 | 90.2% |

## 2.4 Supplementary Table S4: Head-to-Head Baseline Comparison (Full)

**Table 4:** Remote homology accuracy on identical 500-pair balanced evaluation (`protein_pair_remote`, seed 42). All methods evaluated on the same pair set.

| Method | Type | Acc (default) | Acc (optimal) | ROC AUC |
|---|---|---|---|---|
| ESM-2 650M | PLM (mean-cosine) | 50.50% | 51.80% | 0.513 |
| ESM-2 3B | PLM (mean-cosine) | 50.00% | 51.20% | 0.512 |
| MMseqs2 (easy-search, $s$=7.5) | alignment | 54.40% | 55.60% | 0.556 |
| DIAMOND (more-sensitive) | alignment | 53.20% | 53.20% | 0.532 |
| Gemma-4-Instruct (vocab-ext, ZS) | LM (chat) | 60.00% | — | — |
| OmniGene-4 v3 | LM (chat) | 59.50% | — | — |
| OmniGene-4 v4 | LM (chat) | 82.00% | — | — |
| **OmniGene-4 v5** | LM (dual-head) | **82.60%** | — | — |

## 2.5 Supplementary Table S5: Multi-task Generation Evaluation (v5)

**Table 5:** Multi-task generation accuracy across six evaluation categories, 100 samples each, on a held-out split. Keyword-overlap is the standard text-overlap score; Structure additionally reports character-level overlap on the 3Di/DSSP output strings.

| Category | Keyword score | Char overlap |
|---|---|---|
| Protein | 1.000 | — |
| Mol | 0.980 | — |
| Cell | 0.966 | — |
| Literature | 0.559 | — |
| Mutation | 0.113 | — |
| Structure (gen mode) | 0.000 | 0.259 |

## 2.6 Supplementary Table S6: Per-residue Classification Head Accuracy (v5)

**Table 6:** Per-residue accuracy of the two classification heads on a 50-protein held-out evaluation, under teacher-forced reference-residue inference.

| Head | Classes | Chance | Accuracy | N proteins |
|---|---|---|---|---|
| 3Di (Foldseek) | 20 | 5.0% | 78.6% | 19 |
| DSSP (secondary str.) | 8 | 12.5% | 100.0% | 31 |

## 2.7 Supplementary Table S7: SFT Hyperparameters Across All Stages

**Table 7:** SFT hyperparameters. v2/v3 used a chat-style template; v4 switched to pure Alpaca with loss masking and task oversampling. v5 adds dual-head classifiers trained jointly with generation.

| Parameter | v2 / v3 | v4 | v5 |
|---|---|---|---|
| Init from | v2 / Gemma-bio | v3 LoRA + embed | v4 LoRA + embed |
| Training data | 179K / 199K instr | 199K (resampled) | ∼21K Structure |
| Prompt template | chat tags | pure Alpaca | pure Alpaca |
| Loss masking | no | yes (−100) | yes (−100) |
| Joint loss | gen CE only | gen CE only | 0.5 gen + 0.5 cls |
| Effective batch | 64 | 64 | 32 |
| Learning rate | 5e-5 / 2e-5 | 2e-5 | 1e-5 |
| Sequence length | 1024 | 1536 | 1536 |
| Steps | 3,118 / 3,118 | 4,257 | 1,350 |
| GPU-hours | 11.8 / 13.2 | ∼30 | ∼5 |

## 2.8 Supplementary Table S8: Case-Study Pair Details

**Table 8:** Five case-study protein pairs. ESM2 columns: binary prediction at cosine threshold 0.5. MMseqs/-DIAMOND: 1 if hit found at E$\leq 10^{-3}$, 0 otherwise. v5: parsed answer. Y = correct, N = wrong.

| Case | Type | Len1 | Len2 | Label | ESM-650 | ESM-3B | MMseqs | DIAM | v5 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | A | 107 | 106 | 1 | 1(Y) | 1(Y) | 0(N) | 0(N) | 1(Y) |
| 2 | A | 142 | 161 | 1 | 1(Y) | 1(Y) | 0(N) | 0(N) | 1(Y) |
| 3 | B | 244 | 157 | 0 | 1(N) | 1(N) | 0(Y) | 0(Y) | 0(Y) |
| 4 | C | 143 | 101 | 1 | 1(Y) | 1(Y) | 1(Y) | 1(Y) | 1(Y) |
| 5 | D | 108 | 176 | 1 | 1(Y) | 1(Y) | 0(N) | 0(N) | 0(N) |

### 2.9 Supplementary Table S9: Layer-12 Top-3 Routed Experts per Case

**Table 9:** Top-3 routed experts at layer 12 for each case-study pair. The same expert set $\{E_{50}, E_9, E_{108}\}$ activates on all five cases, supporting the conclusion that homology decisions occur inside expert computation rather than at the routing gate.

| Case | Outcome | Top-1 | Top-2 | Top-3 |
|---|---|---|---|---|
| 1 | v5 correct | E50 | E9 | E39 |
| 2 | v5 correct | E50 | E9 | E108 |
| 3 | v5 correct | E50 | E9 | E108 |
| 4 | v5 correct | E50 | E108 | E9 |
| 5 | v5 wrong | E50 | E108 | E9 |

Layer-12 entropy is stable across cases ($H = 2.70$–$2.83$), and per-layer JS divergence between any homologous case and the non-homologous Case 3 stays below 0.04 across all 30 layers (peak at layer 5, 0.040). For comparison, the layer-averaged cross-task JS is 0.230 (Table S1). The $>5\times$ difference supports the gate-vs-expert interpretation in the main text.

### 2.10 Supplementary Table S10: Per-Task Entropy Analysis

**Table 10:** Shannon entropy of expert distributions at layer 12 across training stages. Lower entropy = more concentrated expert use.

| Task | Baseline | CPT | v3 (CPT+SFT) |
|---|---|---|---|
| NL | 3.85 | 4.13 | 4.15 |
| DNA | 2.89 | 2.76 | 2.78 |
| Protein | 2.76 | 2.54 | 2.56 |
| StdHomology | 2.68 | 2.48 | 2.50 |
| RemHomology | 2.71 | 2.52 | 2.54 |
| Cell | 2.95 | 2.88 | 2.90 |
| Mol | 3.02 | 2.95 | 2.97 |
| Structure | 2.82 | 2.65 | 2.67 |

### 2.11 Supplementary Table S11: Token-Level Expert Specialization at Layer 12

**Table 11:** Top-5 tokens for selected experts at layer 12 (v3 model). Purity = fraction of that expert's activations coming from the dominant modality.

| Expert | Purity | Top-5 Tokens |
|---|---|---|
| E54 | 80% | `the`, `of`, `and`, `to`, `in` |
| E59 | 46% | `AA`, `AT`, `TT`, `GG`, `CC` |
| E78 | 35% | `M`, `L`, `V`, `I`, `F` |
| E97 | 28% | `CD4`, `CD8`, `IL`, `TNF`, `IFN` |
| E100 | 15% | `(`, `)`, `=`, `C`, `O` |

# 3 Supplementary Figures

## 3.1 Supplementary Figure S1: Per-Layer JS Curves

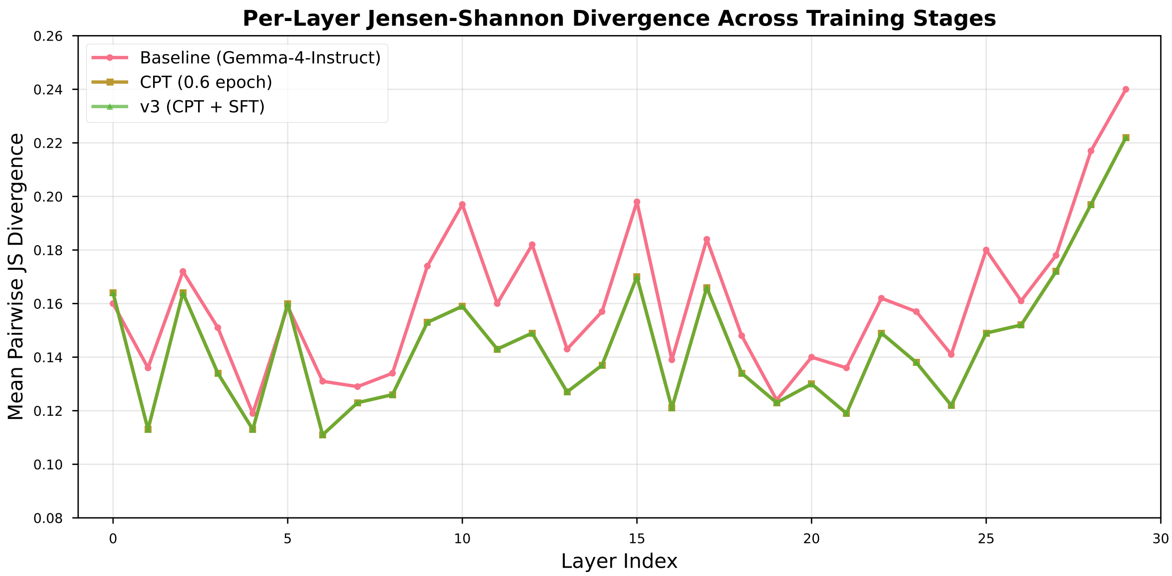


**Figure 1:** Layer-by-layer Jensen–Shannon divergence across training stages. CPT shows the largest increase in middle layers ($L_{11}$–$L_{22}$), while SFT shows concentrated changes near the output layers ($L_{28}$–$L_{29}$).

## 3.2 Supplementary Figure S2: Expert Activation Heatmap

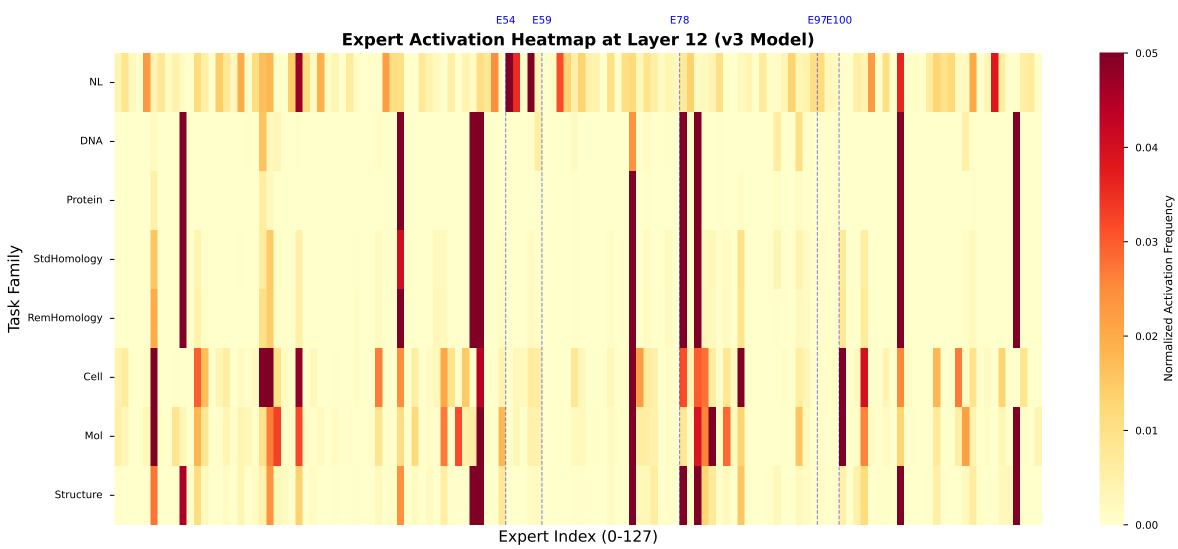


**Figure 2:** Expert activation heatmap at layer 12 for eight task families. Rows: tasks. Columns: 128 experts. Color intensity indicates normalized activation frequency. Blue dashed lines mark key experts (E54: English function words, E59: DNA 2-mers, E78: amino acids, E97: cell markers, E100: SMILES punctuation).

### 3.3 Supplementary Figure S3: Bootstrap Distribution

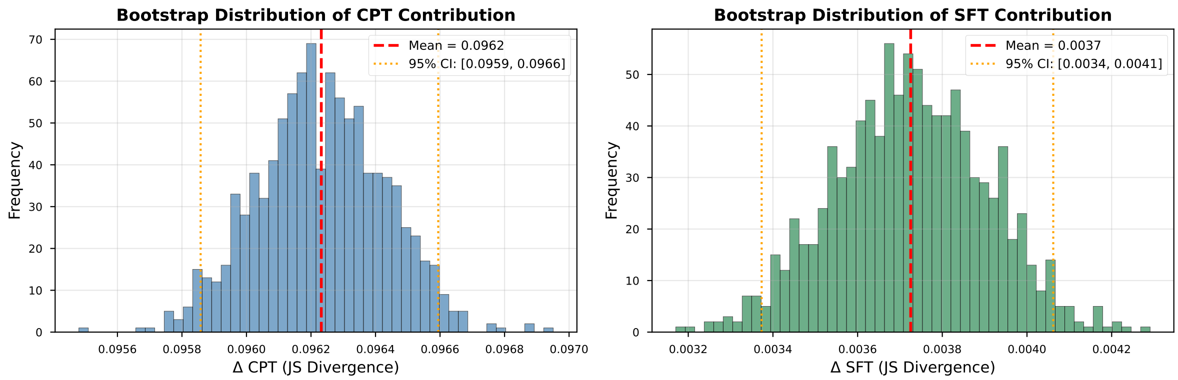


**Figure 3:** Bootstrap distribution of CPT and SFT contributions to JS divergence (1,000 iterations). Both distributions are narrow and exclude zero. Left: $\Delta$ CPT = 0.096 [0.096, 0.097]. Right: $\Delta$ SFT = 0.004 [0.003, 0.004].

### 3.4 Supplementary Figure S4: Case-Study Routing Mosaic

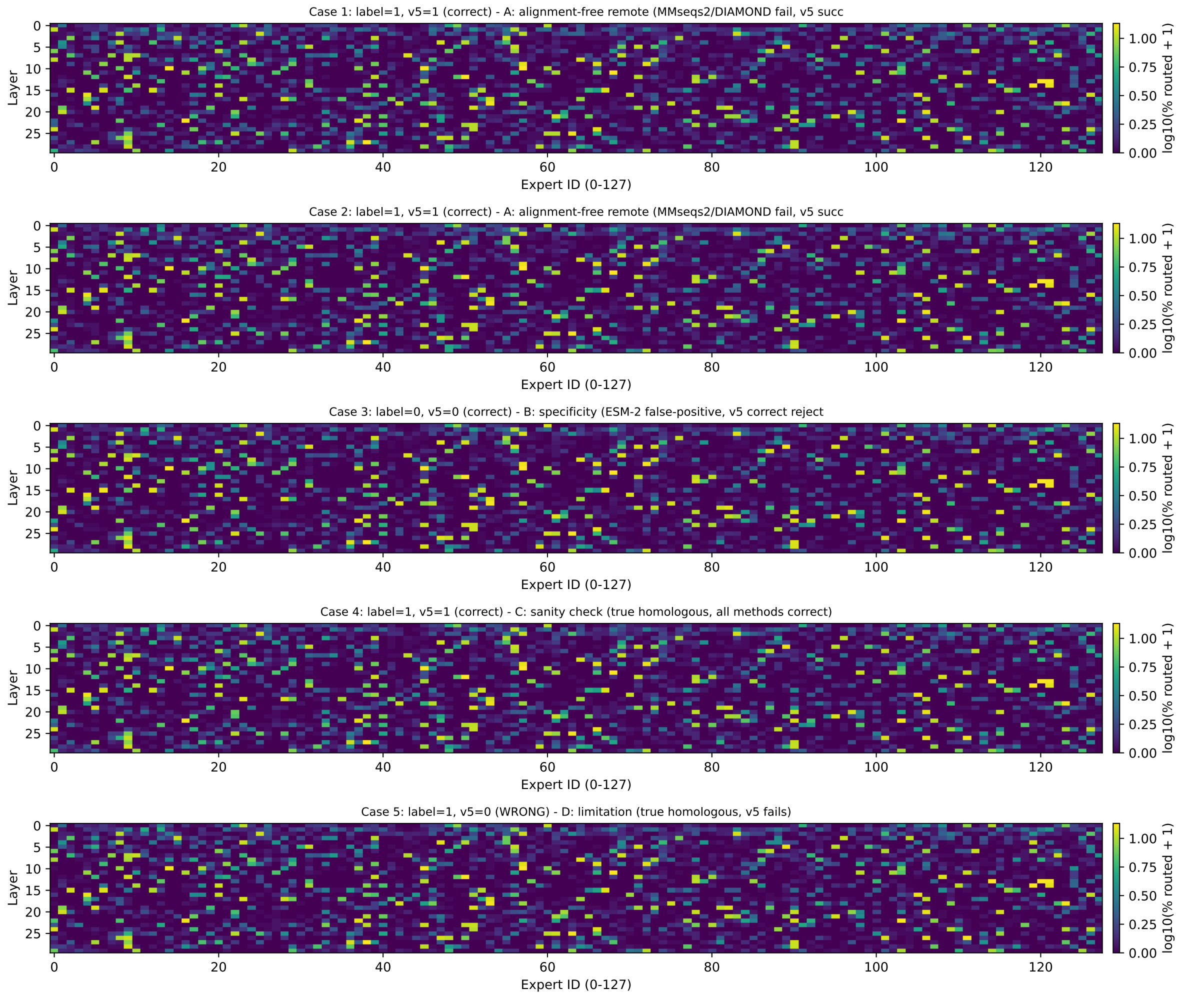


**Figure 4:** Per-case routing-fraction matrix (30 layers × 128 experts), log-scaled. Visually similar across all five cases, with the strongest activation cluster near experts 0–20 and 100–115 in middle layers. Consistent with the protein-rich expert population identified in Table S11.

# 4 Code and Model Availability

All code, training scripts, and analysis notebooks are available at:

`https://github.com/maris205/omnigene4`

## 4.1 Key Scripts

- `biopaws/cpt/1-prepare_cpt_data_mp.py` – CPT data preparation
- `biopaws/cpt/2-run_cpt.py` – CPT training (8-GPU DDP)
- `biopaws/sft_data/17-train_bio_sft_v3_remote.py` – SFT v3
- `biopaws/cpt/40-train_bio_sft_v4.py` – SFT v4 (Alpaca + loss-mask)

- `biopaws/cpt/41-train_bio_sft_v5_classifier.py` – SFT v5 (dual-head)
- `biopaws/cpt/43-eval_v5_full.py` – v5 evaluation
- `biopaws/cpt/44-merge_v5_to_full.py` – v5 LoRA → BF16 merge
- `biopaws/cpt/50-esm2_3b_remote_500pair.py` – ESM-2 3B baseline
- `biopaws/cpt/51-classical_baselines_500pair.py` – MMseqs2/DIAMOND
- `biopaws/cpt/52-collect_per_pair_predictions.py` – Per-pair collection
- `biopaws/cpt/53-pick_case_studies.py` – Case-study selection
- `biopaws/cpt/54-extract_routing_heatmap.py` – Router hook extraction
- `biopaws/cpt/55-plot_case_studies.py` – Figure generation

### 4.2 Model Weights (Hugging Face)

- `https://huggingface.co/dnagpt/OmniGene-4-CPT-v2-merged` – CPT-only, BF16 (∼50 GB)
- `https://huggingface.co/dnagpt/OmniGene-4-SFT-v3-merged` – v3, BF16 (∼50 GB)
- `https://huggingface.co/dnagpt/OmniGene-4-SFT-v3-GGUF` – v3, Q4_K_M (∼16 GB)
- `https://huggingface.co/dnagpt/OmniGene-4-SFT-v4` – v4 LoRA (∼1.9 GB)
- `https://huggingface.co/dnagpt/OmniGene-4-SFT-v5` – v5 LoRA + heads (∼1.9 GB)
- `https://huggingface.co/dnagpt/OmniGene-4-SFT-v5-merged` – v5, BF16 + heads (∼52 GB)
- `https://huggingface.co/dnagpt/OmniGene-4-MM-LoRA` – **MM v3 LoRA + extended embedding** (∼1.7 GB)
- `https://huggingface.co/dnagpt/OmniGene-4-MM-merged` – **MM v3, BF16, stand-alone** (∼49 GB)

## 5 Supplementary Note 2: OmniGene-4-MM Reproducibility

The multi-modal extension reported in Section 4 of the main text was trained in three sequential LoRA stages on top of the v5-merged checkpoint, on a single H20 GPU, in roughly 1.5 GPU-days total wall-clock. All scripts referenced below live under `omnigene5/scripts/` in the project repository.

### 5.1 Pipeline Scripts

- `omnigene5/scripts/01-build_unified_jsonl.py` – Assemble a single multi-modal training corpus from Vis-CheBI20, PubMedVision, HPA10M, ChartQA, BiomedVis, and the v5 SFT pool.
- `omnigene5/scripts/02-fix_image_paths.py` – Resolve relative image paths after the corpus assembly step.

- `omnigene5/scripts/30-train_omnigene5_stage1.py` – **Stage 1** (vision warmup, $\sim$ 10K steps, LR $5\times10^{-5}$).
- `omnigene5/scripts/50-train_stage2.py` – **Stage 2** (mixed text + vision, 6K steps, LR $5\times10^{-6}$).
- `omnigene5/scripts/82-train_stage3v2.py` – Stage 3 v2 (failed: trainable embedding plateaued at 59% standard homology; kept for reproducibility of the failure mode).
- `omnigene5/scripts/83-train_stage3v3.py` – **Stage 3 v3 (final)** (LR $2\times10^{-5}$, frozen embedding, 3000 steps; produces the released checkpoint).
- `omnigene5/scripts/60-eval_stage2.py` – Stage 2 evaluation.
- `omnigene5/scripts/91-eval_stage3v2.py` – Stage 3 v2 evaluation.
- `omnigene5/scripts/92-eval_stage3v3.py` – **Stage 3 v3 evaluation** (BioPAWS standard + remote, Vis-CheBI20 five subtasks, multi-task generation).
- `omnigene5/scripts/95-qualitative_demo_stage3v3.py` – Qualitative-showcase generation, 39 examples (vision + homology + multi-task), used for Figure 8.
- `omnigene5/scripts/71-router_analysis_mm_extended.py` – 8-modality router-level analysis on the Stage 2 checkpoint, used for the modality-invariant-transfer evidence in Section 4.
- `omnigene5/scripts/98-merge_mm_v3.py` – Merge MM v3 LoRA + extended embedding into the v5-merged base, producing the stand-alone BF16 checkpoint released as `dnagpt/OmniGene-4-MM-merged`.

## 5.2 Stage 3 v3 Hyperparameters (Final Checkpoint)

**Table 12:** Stage 3 v3 training hyperparameters.

| Setting | Value |
|---|---|
| Initialization | v5 LoRA + extended embedding (Stage 2 checkpoint) |
| Adapter name | `stage2` (matches Stage 2's saved state-dict keys) |
| LoRA rank / $\alpha$ | $r$=64, $\alpha$=128, dropout 0.05 |
| Target modules | `q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj, router.proj` |
| Embedding | **Frozen** (preserves the v5 paraphrase-transfer signal) |
| Learning rate | $2\times10^{-5}$ (cosine, 3% warmup) |
| Steps | 3,000 |
| Effective batch size | 32 ($1 \times 32$ gradient-accumulation) |
| Max sequence length | 1,536 |
| Optimizer | paged-AdamW-8bit |
| Precision | BF16 |
| Gradient checkpointing | on (non-reentrant) |
| Data | 45K text (homology oversampled $\times 3$) + 5K vision replay |
| Hardware | $1 \times$ NVIDIA H20 (80 GB) |
| Wall-clock | $\sim$ 12 hours |

### 5.3 Per-Stage Capability Trajectory

Table 13 reports the evaluation numbers underlying Figure 7. Stage 3 v2 plateaued at the Stage 2 level despite running for 6K steps; freezing the embedding and raising the learning rate 4× in Stage 3 v3 broke the plateau on the first attempt.

**Table 13:** OmniGene-4-MM capability across all training stages. Vision metrics are accuracy on a 50-prompt-per-task subset of Vis-CheBI20; multi-task generation is keyword-overlap on a 30-prompt-per-category subset of an internal SFT-eval set; homology is BioPAWS standard / remote.

| | **Vision (Vis-CheBI20)** | | | | | **Multi-task gen** | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Stage** | rec. | cap. | desp | IUPAC | SMILES | Cell | Mol | Prot. | Lit. | **Hom. std/rem** |
| v5 (text only) | — | — | — | — | — | — | — | — | — | 99.4 / 82.6 |
| MM Stage 1 | 1.00 | 0.90 | 0.10 | 0.00 | 0.00 | — | — | — | — | —* |
| MM Stage 2 | 1.00 | 0.88 | 0.14 | 0.00 | 0.00 | 0.25 | 0.32 | 1.00 | 0.17 | 59.0 / 56.5 |
| MM Stage 3 v2 | 1.00 | 0.90 | 0.16 | 0.00 | 0.00 | 0.75 | 0.79 | 1.00 | 0.24 | 59.5 / 60.0 |
| **MM Stage 3 v3** | **1.00** | **0.96** | 0.14 | 0.00 | 0.00 | **0.95** | **0.91** | **1.00** | **0.30** | **85.0 / 69.5** |

* Stage 1 produces parseable BioPAWS responses on only 4/200 prompts (catastrophic forgetting), so a homology number is not meaningful.

### 5.4 8-Modality Routing Analysis

The 8-modality routing-divergence numbers reported in the main text are collected by installing `forward_hooks` on all 30 `Gemma4TextRouter` modules of the Stage 2 checkpoint, running 50 prompts per modality across the eight modality categories (*vision_molecule*, *vision_medical*, *vision_pathology*, *vision_chart*, *protein_sequence*, *dna_sequence*, *structure_3di*, *natural_language*), and computing pairwise Jensen–Shannon divergence per layer over the resulting $30 \times 128$ expert-count matrices, then averaging over layers. The full $8 \times 8$ divergence matrix and per-layer curves are saved to `outputs/OmniGene-4-MM-stage2/router_analysis_8mod/` when the analysis script is re-run; key numbers are reproduced below.

**Table 14:** Selected pairwise routing JS divergences from the 8-modality analysis (layer-averaged, Stage 2 checkpoint).

| **Pair** | **Layer-averaged JS** |
|---|---|
| *Within sequence cluster* | |
| protein_sequence vs structure_3di | 0.016 |
| protein_sequence vs dna_sequence | 0.038 |
| *Within vision cluster* | |
| vision_medical vs vision_pathology | 0.048 |
| vision_molecule vs vision_chart | 0.072 |
| *Cross-cluster (representative)* | |
| vision_pathology vs natural_language | 1.21 |
| vision_molecule vs protein_sequence | 1.24 |
| dna_sequence vs natural_language | 1.31 |

The within-cluster pairs all sit below 0.08, while cross-cluster pairs all exceed 1.2 – a clean three-tier separation that emerges with no modality supervision (the only training signal is token-level cross-entropy). This is the empirical evidence supporting the modality-invariant-transfer claim in the main text.

## 5.5 8-Modality Routing Figures

Figure 5 shows the full $8 \times 8$ pairwise JS divergence matrix; Figure 6 shows the modality-by-expert routing distribution at layer 12; Figure 7 plots the per-layer JS curves for representative cross-cluster and within-cluster pairs.

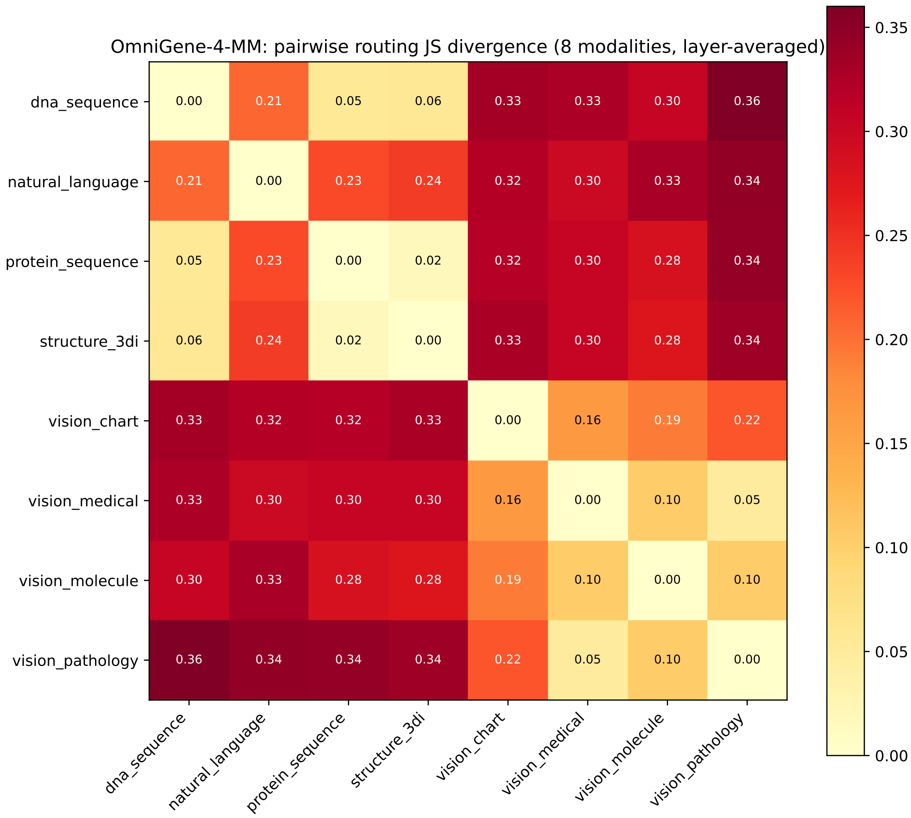


**Figure 5:** Pairwise routing JS divergence across the eight modalities (layer-averaged, Stage 2 checkpoint). Within-cluster cells are dark (low divergence), cross-cluster cells are bright (high divergence).

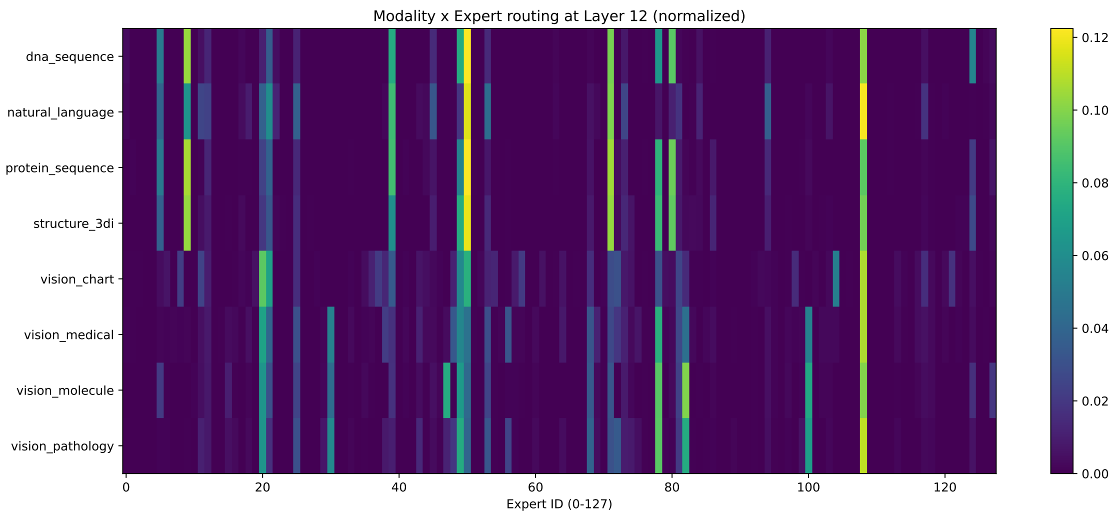


**Figure 6:** Modality × expert routing at layer 12 (peak cross-modality differentiation), normalized within each row. Each row is one of the eight modalities; each column is one of the 128 experts.

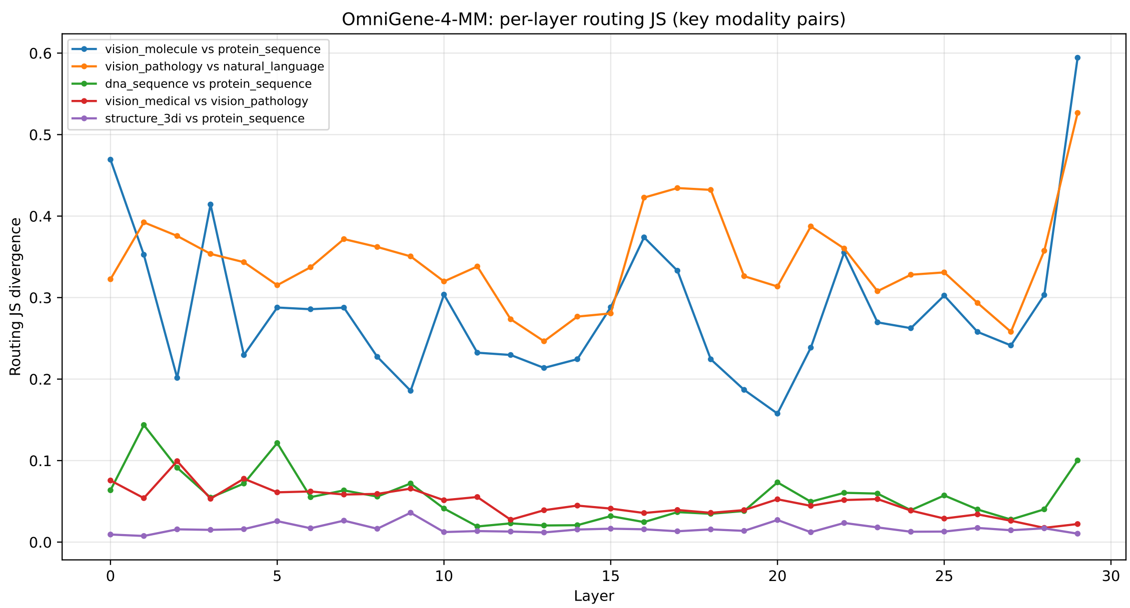


**Figure 7:** Per-layer routing JS divergence for selected modality pairs across the 30 transformer layers. Cross-cluster pairs (e.g. *vision_pathology* vs *natural_language*) sit consistently above 1.2; within-cluster pairs (e.g. *protein_sequence* vs *structure_3di*) sit below 0.05 across all layers.

# 6 Supplementary Note 3: Released Training Data

To support reproducibility, all training corpora used in the paper — both for OmniGene-4 (CPT + Bio-SFT v2–v5) and for the OmniGene-4-MM extension (Stage 1–3 LoRA) — are released as four publicly accessible Hugging Face dataset repositories under the `dnagpt` organization. Reviewers and practitioners can clone or stream any of them without authentication. The top-level repository layout is shown below.

**Table 15:** Released Hugging Face dataset repositories supporting the OmniGene-4 / OmniGene-4-MM training pipeline.

| Repository | Total size | Used by |
|---|---|---|
| `dnagpt/omnigene4-cpt-corpus` | $\approx 96$ GB | CPT (§??) |
| `dnagpt/omnigene4-sft-data` | $\approx 700$ MB | Bio-SFT v2–v5; OmniGene-4-MM Stage 2/3 |
| `dnagpt/omnigene4-mm-corpus` | $\approx 500$ MB | OmniGene-4-MM Stage 1–3 |
| `dnagpt/biopaws` | $\approx 50$ MB | BioPAWS evaluation (standard + remote homology) |

## S3.1: `dnagpt/omnigene4-cpt-corpus` (CPT corpus, $\approx 96$ GB)

The full continued-pre-training mixture used for OmniGene-4. The 1:1:1 biological-to-natural-language ratio described in the main text is realized by the file sizes below.

| File | Size | Source / Description |
|---|---|---|
| `dna_32g.txt` | 31 GB | DNA sequences sampled from public genomes (one sequence per line) |
| `protein_uni_16.txt` | 16 GB | UniRef-derived protein amino-acid sequences |
| `protein_lucaone_15g.txt` | 15 GB | Protein sequences sampled from the LucaOne pretraining pool |
| `openwebtext.txt` | 37 GB | English text replay split (sampled from `Skylion007/openwebtext`); kept here so t |
| `pdb_aa.fasta` | 100 MB | Paired PDB amino-acid sequences (paired with `pdb_3di.fasta` for the 3Di predic |
| `pdb_3di.fasta` | 100 MB | Paired PDB Foldseek-3Di sequences |
| `ss.txt` | 263 MB | DSSP secondary-structure labels (per residue) |

## S3.2: `dnagpt/omnigene4-sft-data` (SFT corpus, $\approx 700$ MB)

Cumulative supervised-fine-tuning corpus consumed by Bio-SFT v2 through v5 and re-used by OmniGene-4-MM Stage 2/3. The most important file is `train/omnigene_sft_v1_train_with_remote.jsonl`, which is the 199K-row corpus used by every model from v3 onward.

| File | Rows | Used by |
|---|---:|---|
| `bio_sft_v2_train.jsonl` | $\sim 179$K | Bio-SFT v2 (eight task families) |
| `distill_seed.jsonl` | $\sim 6$K | Initial distillation seed (predecessor of v2) |
| `train/omnigene_sft_v1_train.jsonl` | $\sim 179$K | SFT v3 base (cleaner schema, Alpaca template) |
| **`train/omnigene_sft_v1_train_with_remote.jsonl`** | $\sim 199$K | **Final SFT corpus used by Bio-SFT v3, v4, v5, a** |
| `eval/omnigene_sft_v1_eval.jsonl` | $\sim 1.5$K | Held-out evaluation split (used by all evaluation scripts |
| `master/omnigene_sft_v1_master.jsonl` | $\sim 285$K | Pre-split master corpus (intermediate; not used in train |
| `master/cell_sft_master.jsonl` | $\sim 37$K | Cell-biology task subset |
| `master/mol_sft_master.jsonl` | $\sim 50$K | Molecule task subset |
| `stats/data_mix_report.json` | – | Per-category counts and mix ratios |

Each row is a JSON object with at least the fields `instruction`, `input`, `output`, and `category`; some rows additionally carry `task_name` and `subtask` for fine-grained accounting.

## S3.3: `dnagpt/omnigene4-mm-corpus` (MM corpus, $\approx 500$ MB)

Multi-modal training corpus used by OmniGene-4-MM Stage 1 (vision warmup), Stage 2 (mixed text + vision), and Stage 3 (homology specialty with vision replay).

| File | Rows | Notes |
|---|---|---|
| train.jsonl | $\sim$ 280K | Unified multi-modal training file (vision + text) used by Stage 1 directly and as a po |
| val.jsonl | $\sim$ 5K | Held-out validation slice |
| meta.json | – | Per-modality counts and split sizes |
| B_chebi20/train.json | $\sim$ 26K | Vis-CheBI20 train split (project copy of PharMolix/Vis-CheBI20). Five OCSU subta |
| B_chebi20/test.json | $\sim$ 3K | Vis-CheBI20 test split (same subtasks); the source of every Vis-CheBI20 number repo |

The unified `train.jsonl` carries each row's `messages` (chat-format), `images` (list of relative paths), and `modality` fields. Image rows reference local paths under the project's data tree; reviewers reproducing the MM training can either point those paths at their own copies of the upstream image datasets, or run `omnigene5/scripts/02-fix_image_paths.py` to remap.

### S3.4: `dnagpt/biopaws` (BioPAWS evaluation, $\approx 50\,\mathrm{MB}$)

The BioPAWS protein-pair benchmark used for all standard and remote homology numbers reported in the main text. This dataset was first released as part of the BioPAWS paper (Wang, 2026).

- `protein_pair_short`: close-homolog pairs (the "Standard" homology benchmark).
- `protein_pair_remote`: structurally related but sequence-divergent pairs (the "Remote" homology benchmark, the headline metric of the paper).

For the v3 SFT recipe (§**??**) we draw 20,000 `protein_pair_remote` rows for training (10K label-1 + 10K label-0); the 2,000 evaluation rows are held out by deterministic seed-42 sampling and never appear in the SFT corpus.

## Loading Pattern

A minimal load + run pattern, end-to-end (model + data, no authentication required for either):

```
from transformers import AutoTokenizer, AutoProcessor, AutoModelForCausalLM
from datasets import load_dataset
import torch

# 1. Load the merged multi-modal model in one call
REPO = "dnagpt/OmniGene-4-MM-merged"
tok   = AutoTokenizer.from_pretrained(REPO)
proc  = AutoProcessor.from_pretrained(REPO)
model = AutoModelForCausalLM.from_pretrained(
    REPO, torch_dtype=torch.bfloat16, device_map="auto"
).eval()

# 2. Stream the SFT corpus
sft = load_dataset("dnagpt/omnigene4-sft-data",
                   data_files="train/omnigene_sft_v1_train_with_remote.jsonl",
                   split="train", streaming=True)
print(next(iter(sft)))

# 3. Stream the MM corpus
```

```
mm = load_dataset("dnagpt/omnigene4-mm-corpus",
                  data_files="train.jsonl",
                  split="train", streaming=True)
print(next(iter(mm)))
```